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Integrating In Situ, Remote Sensing, And Physically Based Modeling Approaches to Understand Global Freshwater Ice Dynamics

Key Points:

- Monthly lake-wide evaporation is not strongly correlated with ice cover
- Ice impact on evaporation is less than the uncertainty in evaporation estimates themselves
- Winter large lake evaporation is governed by overall meteorologic conditions

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Does Ice Cover Cap Evaporation in Large Lakes?

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Abstract Lakes are experiencing ice declines and fundamental changes in winter conditions. For Earth's largest lakes that experience seasonal ice cover, the relationship between ice conditions and evaporation is critical to water balance estimates and global freshwater storage. Here, we analyze robust data sets of net basin supplies, satellite-derived products, and model estimates of surface turbulent heat flux for the Laurentian Great Lakes during the period 1973–2022. We show that ice cover does not have a strong relationship with lake evaporation in winter months and that often the magnitude of the ice effect on moisture flux reduction is within the range of natural variability and the uncertainty of water budget estimates. This suggests that differences in lake evaporation between cold and warm winters is driven by seasonal overlake atmospheric conditions, more broadly, and that ice cover reduces but does not determine the resultant evaporation.

Plain Language Summary Large lakes hold most of the planet's surface freshwater. As winters become shorter and ice cover decreases due to climate change, understanding how ice affects lake evaporation is critical for managing freshwater supplies. This study analyzes 50 years of data on the North American Great Lakes to examine the relationship between ice cover and evaporation during winter months (January–March). The findings reveal that while ice cover reduces evaporation locally, it is not the primary driver of lake-wide winter evaporation. Instead, evaporation is more influenced by other atmospheric factors, such as wind speed, humidity, and temperature differences above the lake. Only in extreme cases, when over 90% of the lake surface is covered in ice, would the sudden absence of ice significantly increase evaporation. These results challenge the assumption that ice strongly caps evaporation and emphasize the role of broader atmospheric conditions. The findings are vital for predicting how freshwater resources will respond to future climate conditions and for managing water supplies in Earth's largest lakes.

1. Introduction

Earth's largest lakes (>15,000 km²) are responsible for the majority of the liquid surface freshwater on the planet (Shiklomanov, 1993). However, for many of these lakes, the closure of the water budget remains an unsolved or difficult problem (Gronewold et al., 2020). While large lakes play a significant role in water resources management and weather and climate prediction, our understanding of how they are responding to climate change remains limited (Anderson et al., 2021, 2024; Gronewold et al., 2013; Moore et al., 2009; Piccolroaz & Toffolon, 2018; Rao et al., 2012; Rühland et al., 2023). In particular, as large lakes have experienced loss of ice and shortening of the winter season, the complex relationship between changing winter conditions and water budget components are top of mind for predicting how Earth's freshwater supply will respond to future conditions (Anderson et al., 2024; Lin et al., 2022; Ozersky et al., 2021; J. Wang et al., 2017, 2020).

One such complexity is the relationship between ice coverage and lake evaporation (Blanken et al., 2011; Sharma et al., 2019; Spence et al., 2011; Van Cleave, 2012; Yao et al., 2023; Zhao et al., 2022). In particular, the ice-albedo feedback has been identified as a significant driver of change to lake thermal structure (Austin & Colman, 2007; Sharma et al., 2019; W. Wang et al., 2018). While it has been noted that high fall evaporation rates can drive increased winter ice formation (Van Cleave, 2012), and lake evaporation also depends on other atmospheric factors such as wind and vapor pressure (Spence et al., 2013), it is often thought that, when formed, ice cover effectively shuts down turbulent surface fluxes between the lake and the atmosphere and therefore acts as a deterrent to evaporation and lake-effect precipitation (Austin et al., 2025; Bai et al., 2015; J. Wang et al., 2010).

Here, we aim to answer the question, does ice cap evaporation in Earth's largest lakes? The answer to which would inform us about water balance components that are critical for a significant portion of global freshwater storage. Furthermore, it would provide a basis for understanding how the impact of climate change on lake ice conditions

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would in turn affect net basin supply and water resource management. To determine the relationship between ice and evaporation in large lakes, we take two approaches. The first being a practical look at the historical record to see how winter evaporation scales with ice concentration and other environmental conditions. The second being an analysis of evaporation if lake ice were to suddenly be removed. While unrealistic, calculation of the ice-free flux serves as an extreme limit from which we can investigate the direct mechanism of ice as a cap on lake evaporation. For relatively lower ice concentration periods, for example, <50%, this approach may represent realistic conditions of ice fractioning or breakup, particularly in the context of a warming world. For instance, a possible result of climate change could be thinner and weaker ice that is prone to breakup and increased lateral melt (Culpepper et al., 2024; Hampton et al., 2024). Therefore, this analysis also gives insight into how large lake water budgets would be expected to respond to the mechanical and structural repercussions of climate change impacts on lake ice.

This two-prong approach requires a robust understanding of the lake water balance components on time scales such that relationships can be revealed. More specifically, a long-term record of accurate and consistent data at scales capable of resolving the moisture flux response relative to ice cover is paramount. Furthermore, this analysis requires atmospheric and lake data at high temporal and spatial resolutions along with validated models capable of resolving turbulent surface fluxes and ice concentrations. For these reasons, our analysis focuses on the Laurentian Great Lakes, hereafter Great Lakes, which represent 5 of Earth's largest lakes and 21% of its surface freshwater. Recent work on the Great Lakes has yielded long-term, daily records of evaporation (O'Brien et al., 2024), and a set of validated models for lake hydrodynamics, ice conditions, and turbulent heat fluxes (Anderson et al., 2018; Charusombat et al., 2018). In addition, spatially varying, daily ice records are available for the Great Lakes since 1973, enabling a thorough investigation of ice and evaporation relationships.

2. Methods

We assess the relationship between ice cover and evaporation for Earth's largest lakes by analyzing robust data sets of net basin supplies, satellited-derived products, and model estimates of surface turbulent heat flux. We chose the Great Lakes due to the availability of daily evaporation estimates, daily ice concentration data from satellite-informed remote-sensing, and hourly meteorological and lake surface conditions from operational numerical weather (water) prediction (NWP) models. Although, large lakes hold the overwhelming majority of Earth's liquid surface freshwater, this suite of data at high temporal and spatial resolution is not available for other large lakes. We also focus on the Great Lakes because lake-wide ice concentrations have been decreasing since the 1970s, the lakes play an important role in regional weather and climate conditions (Wright et al., 2013), routine reference to lake ice-evaporation relationships are made for these lakes (Austin et al., 2025; J. Wang et al., 2010), and water level variability, and thus variability in a significant portion of Earth's surface freshwater storage, has remained as a point of uncertainty and concern (Gronewold & Stow, 2014; Yao et al., 2023).

We consider ice concentration expressed as a percentage of lake surface area for each of the Great Lakes, lake-wide evaporation estimates, and vapor pressure gradient for total monthly and seasonal values for January, February, and March (JFM) during the period 1973–2022. Although peak evaporation in the Great Lakes occurs in the fall and early winter (O'Brien et al., 2024), notable ice coverage begins in late December or early January and extends through March (Figure 1). We therefore focus our analysis on January–March to capture both the dominant winter ice period and months with the potential for significant ice-reduced evaporation. The characterization of the ice cover and evaporation relationships is produced using data from the entire period. The overlake vapor pressure gradient and evaporation relationships are analyzed for the period of available high-resolution overlake meteorological conditions (1995–2022), and the analysis of hypothetical ice removal impacts on lake evaporation is carried out for the years 2019–2021 using a high-resolution turbulent flux model.

Lake-wide evaporation estimates are obtained from the Large Lake Statistical Water Balance Model (L2SWBM) (Gronewold et al., 2020; O'Brien et al., 2024). The L2SWBM uses a Bayesian framework to combine available hydrological and hydrometeorological data sets and a conventional water balance equation to calculate individual components that close the water budget for large lakes. The L2SWBM is used operationally by the Coordinating Committee on Great Lakes Hydraulic and Hydrologic Data, a bi-national governmental committee tasked with collecting, coordinating, and reporting Great Lakes hydrologic data. Most recently, the L2SWBM was used to generate a 73-year (1950–2022) data set that includes publicly available daily lake-wide evaporation for the Great Lakes (O'Brien et al., 2024). The L2SWBM is the only product, to our knowledge, that is able to close the water

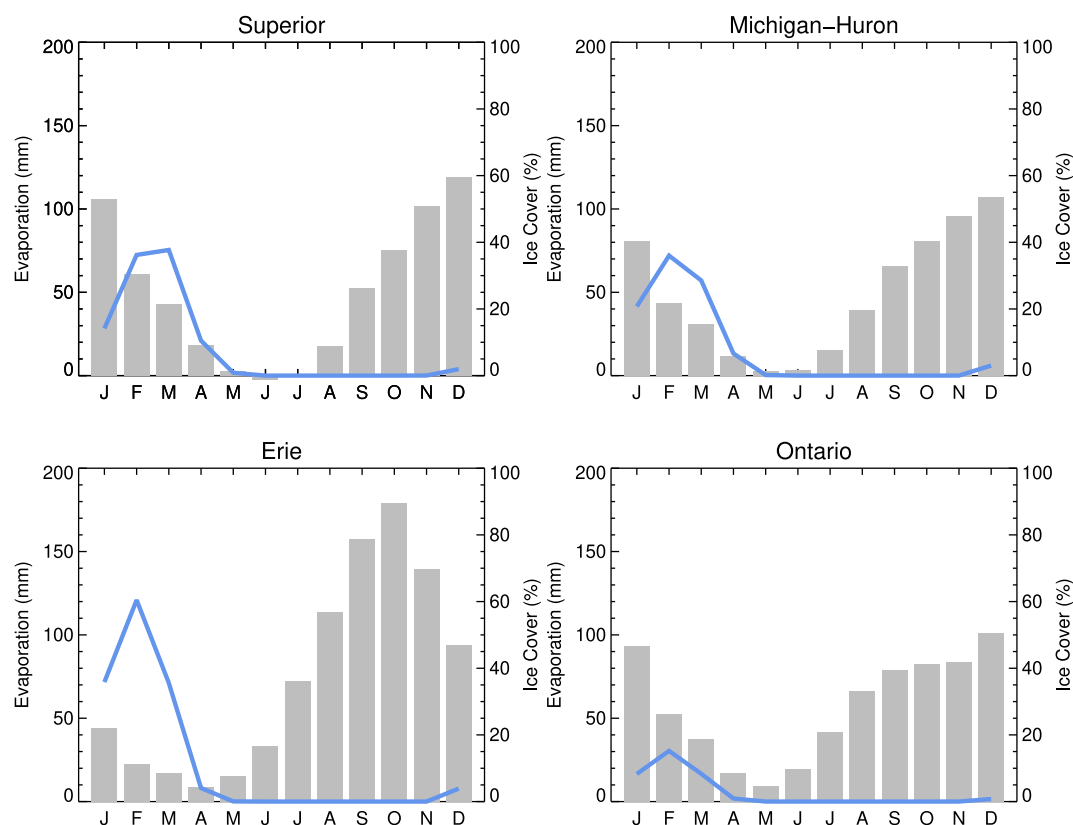


Figure 1. Average monthly evaporation (gray bars) and ice concentration (blue line) for the Great Lakes for 1973–2022.

balance for multi-decadal periods for Earth's largest lakes, and therefore provides the best estimate for lake evaporation from which to carry out this analysis.

We use satellite-derived observations of ice cover from the North American Ice Service (NAIS) for the period 1973–2022 (Yang et al., 2020). The NAIS provides daily georeferenced ice concentrations over the Great Lakes, which are provided on a 1.8 km grid by the National Oceanic Atmospheric Administration (NOAA) Great Lakes Coastwatch Program (NOAA CoastWatch, 2024). The NAIS ice data are created using a suite of observations including satellite remote sensing (e.g., AVHRR, MODIS, GOES, Radarsat-2, etc) and are given in 10% increments representing ice fraction from 0% to 100%. The Coastwatch Program also provides daily lake surface temperatures on the same 1.8 km grid from the Great Lakes Surface Environmental Analysis (GLSEA) since 1995 (Schwab et al., 1999). The GLSEA is derived from observations from the Advanced Clear-sky Processor for Oceans L3S-LEO SST (collated from various satellites) and is widely used for Great Lakes surface temperature analysis and model assessment (Anderson et al., 2018, 2024; Woolway et al., 2021).

Overlake vapor pressure gradients ($e_0 - e_2$) and heat fluxes are calculated from meteorological conditions from the NOAA Great Lakes Coastal Forecast System (GLCFS) (Anderson et al., 2018; Beletsky et al., 2006; Schwab & Bedford, 1994). The GLCFS provides hourly 2-m air temperatures and humidity, downward shortwave and longwave radiation, 10-m wind speed and direction, and air pressure (NOAA Great Lakes Environmental Research Laboratory, 2024). Atmospheric conditions in the GLCFS for the period 1995–2018 are derived from coastal station interpolation (Schwab & Bedford, 1994), and for the period 2019–2022, the GLCFS uses meteorology from the NOAA High-Resolution Rapid Refresh (Benjamin et al., 2016).

We calculate the effect of ice cover on evaporation using the latent heat flux algorithm from the Coupled Ocean-Atmosphere Response Experiment (COARE) (Edson et al., 2013; Fairall et al., 1996, 2003). COARE is a standard approach for computing turbulent surface heat fluxes over the ocean and the Great Lakes (Anderson et al., 2018; Charusombat et al., 2018). For the winter months (January–March) in years 2019–2021, we calculate the hourly latent heat flux over each of the Great Lakes on the 1.8 km grid noted above. Hourly, spatially varying,

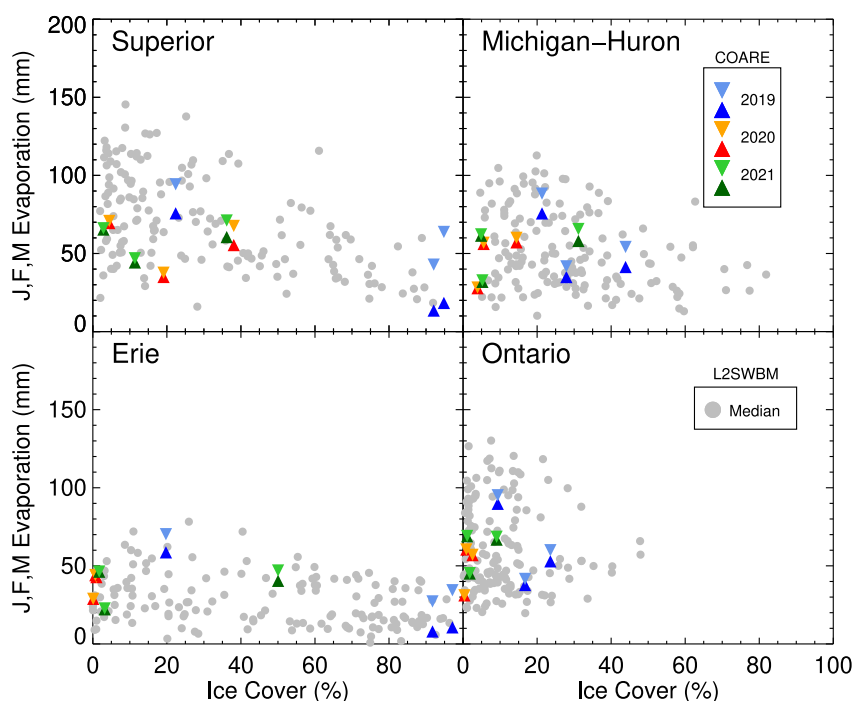


Figure 2. Total monthly evaporation and monthly averaged ice cover for the winter months of January, February, and March for the Great Lakes. Gray circles are the median water balance estimates from L2SWBM for the period 1973–2022. Triangles represent the Coupled Ocean-Atmosphere Response Experiment ice-fractioned ($\triangle$) and unfractioned (∇) evaporation in 2019–2022.

meteorological conditions, ice concentrations, and water temperatures are provided by the GLCFS. Given the modeled ice concentration, we compute two versions of the surface latent heat flux with COARE: (a) using a linear fraction of latent heat flux by ice concentration (following the method of Los Alamos Sea Ice Model (CICE) and GLCFS) (Anderson et al., 2018; Hunke et al., 2017) and (b) an unfractioned latent heat flux that assumes the ice has suddenly been removed. In the latter case, we assume the latent heat flux provided by COARE represents a maximum possible flux under the present environmental conditions. In all cases presented here, the lake is not two-way coupled with the atmosphere, hence there is no feedback from the lake to the atmosphere represented in this analysis. While this is a simplification of the lake flux processes, it allows for the assessment of ice impacts on open water fluxes, which are dominant in evaporation from large lakes with dynamic ice fields. Ultimately, the latent heat flux is converted into daily and monthly values of lake-wide evaporation.

3. Results

Over the entire historical record (1973–2022), the relationship between ice cover and evaporation varies between lakes (Figure 2). Lake Superior is the only lake with a moderate correlation between winter ice cover and lake-wide evaporation (correlation coefficient, $CC = -0.52$). Lakes Michigan-Huron and Erie both have weak correlations ($CC = -0.30$ and -0.31 , respectively), and Lake Ontario, which has no winter months with ice concentrations above 50%, displays the weakest relationship between ice cover and evaporation ($CC = -0.09$). It is important to note that although JFM are the months with the highest ice concentrations for all lakes, total evaporation in these months still constitutes between 26% and 35% of the average annual evaporation, with the exception of Lake Erie, in which these months only make up 9% of the annual total (Figure 1).

Weak relationships between ice cover and lake-wide evaporation is in part due to the role of overlake meteorology in turbulent latent heat fluxes. Vapor pressure gradient alone has weak to moderate correlations with monthly winter evaporation, regardless of ice concentration, where correlation coefficients are 0.41, 0.40, 0.56, and 0.38 for Superior, Michigan-Huron, Erie, and Ontario, respectively (Figure 3). This suggests the additional importance of wind speed and atmospheric stability on surface flux and evaporation.

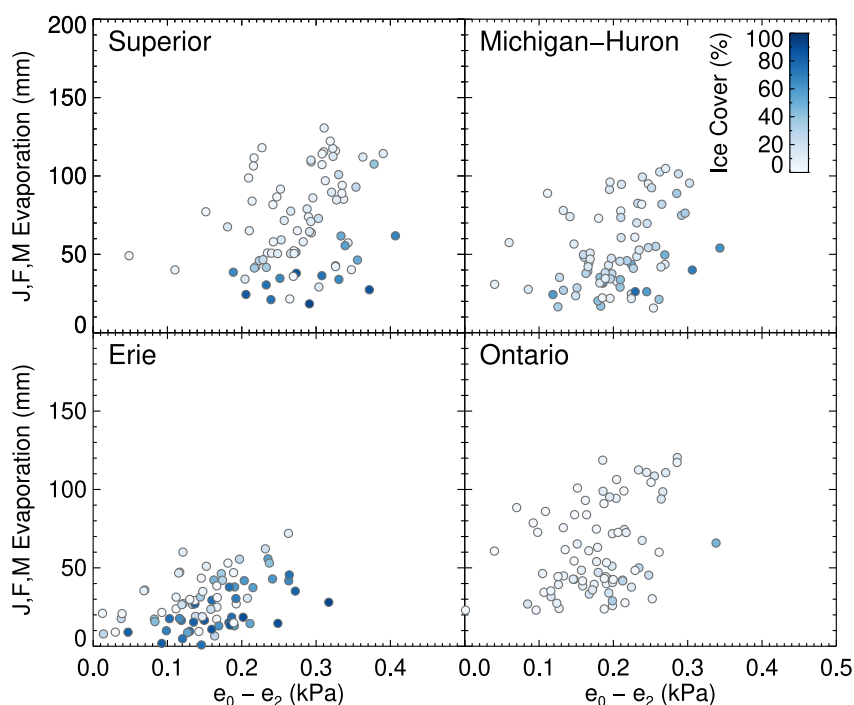


Figure 3. Winter monthly average vapor pressure gradient, $e_0 - e_2$, and L2SWBM evaporation for the period 1995–2022. Values are colored by monthly averaged ice cover.

In 2019, 2020, and 2021, which represent intense, mild, and moderate winters, respectively, ice-free (or unfractionated) latent heat fluxes increase winter evaporation in each month (Figure 2). The amount of increased evaporation is dependent on the ice cover percentage and the baseline latent heat flux derived from overlake atmospheric conditions using COARE. However, in the context of the historical record, nearly all of the ice-free values still fall within the climatological ranges found in the ice-evaporation relationships. The only potential outliers are found in Lake Superior when ice concentrations are greater than 90%, particularly in the case with 95% ice cover, and the resultant ice-free evaporation extends to or just beyond the top of the historical range.

For 2019–2021, water balance estimates of evaporation are highest in January and decrease in February and March (Figure 4). The 2.5 and 97.5 percentiles capture the likely range or uncertainty in the monthly water balance-derived evaporation of L2SWBM. The ranges are fairly consistent across months and across years for each lake, and even across lakes, the differences in range are small with Ontario having the smallest range (10.6 mm) and Superior having the largest (21.7 mm). While ice-fractionated evaporation from the COARE model does not always fall within the L2SWBM range, the COARE values follow the general pattern of flux across months and years. The discrepancies between the COARE model and the L2SWBM estimates may be due to a number of factors including uncertainty in the meteorological conditions, water temperatures, and ice concentrations used in the COARE calculations and structural uncertainty in the COARE algorithm itself. In any case, the impact of ice cover on evaporation is apparent from the difference between the ice-fractionated and unfractionated COARE values (Figures 2 and 4).

The impact of ice cover on evaporation, which is essentially the difference between unfractionated and ice-fractionated latent heat flux from the COARE model, can be put in context of the range from the water balance estimates of evaporation from L2SWBM (Figure 5). We see that the COARE predicted ice impact (ΔE_{COARE}) is in most cases lower than the estimated range of monthly evaporation (ΔE_{L2SWBM}) for these years. When ice concentrations are less than 50%, the difference between the unfractionated and ice-fractionated evaporation is smaller than the uncertainty in the water balance derived evaporation. It is only for extremely high ice concentrations, notably in Lake Superior and Lake Erie when ice cover is greater than 90%, do we see the impact of ice on COARE evaporation rise above the uncertainty range in the statistical water balance model. This suggests that for periods with lower ice concentrations, the ice impact on evaporation falls within the uncertainty in the best estimate of lake evaporation. It

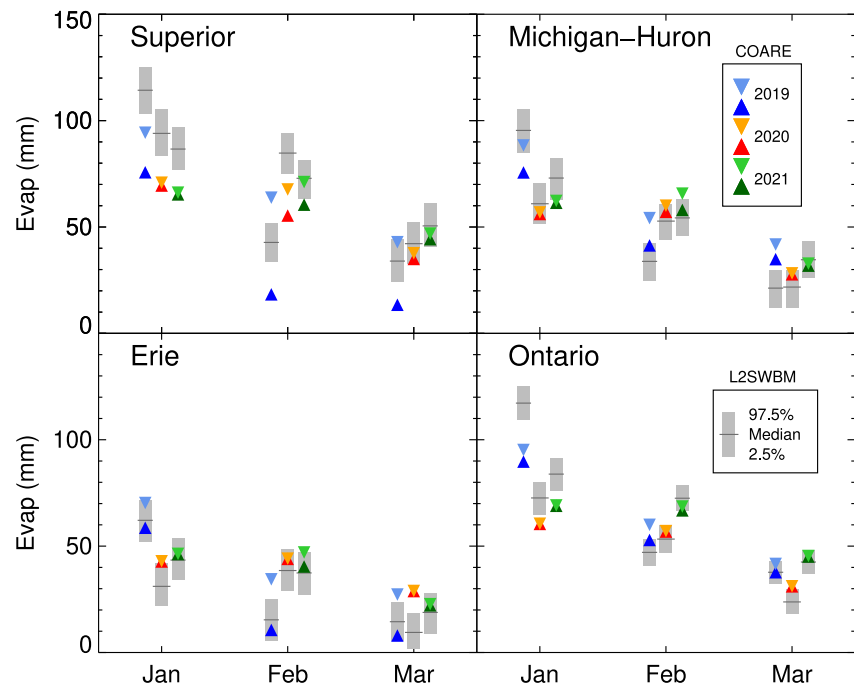


Figure 4. Monthly evaporation estimates from L2SWBM and Coupled Ocean-Atmosphere Response Experiment (COARE) for 2019–2021 (shown left to right for each month). L2SWBM values are shown for median (gray line) and the 2.5 and 97.5 percentile (gray bar) estimates. Triangles represent COARE ice-fractionated ($\triangle$) and unfractionated (∇) evaporation.

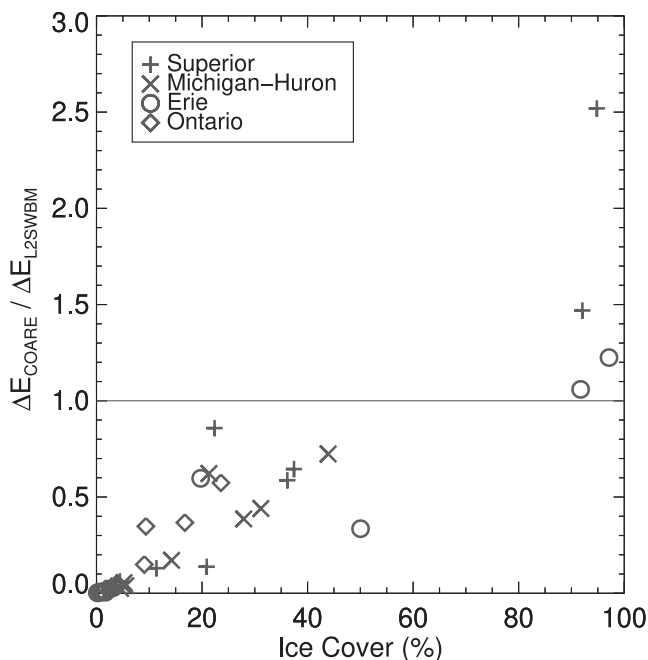


Figure 5. The ratio of the difference between unfractionated and ice-fractionated evaporation from Coupled Ocean-Atmosphere Response Experiment (ΔE_{COARE}) to the evaporation range from L2SWBM (ΔE_{L2SWBM}) as a function of ice cover. Each data point refers to a winter month, January–March, for the years 2019–2021.

is important to note that for these years and winter months, there is a gap in observed ice cover between approximately 50% and 90%. This is consistent with historical trends in Great Lakes ice concentrations, where the individual lakes tend to experience either high or low ice concentration winters. However, it may be reasonable to assume that for winter months with ice cover between 50% and 90% the ice impact on evaporation could be linearly approximated with the data analyzed here (Figure 5).

4. Summary and Conclusions

We have found that the relationship between lake-wide ice cover and evaporation in some of Earth's largest lakes is not as straightforward as suggested in the literature. At the local scale, ice cover reduces lake evaporation from would-be values under otherwise equivalent ice-free conditions. Commonly employed modeling approaches, including those investigated here, represent this as a linear reduction proportional to local ice concentration. However, integrated across a lake-wide scale, the presence of ice cover does not necessarily lead to significantly reduced evaporation.

Historical ice data and net basin supply estimates in the Great Lakes suggest that on a lake-scale, the relationship between ice cover and evaporation is weak for lakes Michigan-Huron, Erie, and Ontario, and only has a moderate correlation in Lake Superior (Figure 2). In addition, with the exception of Lake Erie, evaporation in the winter months of JFM are on par with values in late fall, which is generally the time of greatest lake evaporation, despite these being the months with greatest ice cover (Figure 1). This highlights the importance of atmospheric dynamics such as vapor pressure gradient, stability, and wind speed heterogeneity in surface turbulent heat fluxes.

When simulations of suddenly ice-free conditions are placed in historical context of ice-evaporation relationships, in nearly all cases the increase in winter evaporation is within the range of observed lake-wide evaporation at the corresponding percentage of ice cover (Figure 2). This suggests that while ice cover imposes a mechanistic reduction in local-scale evaporation, the overall relationship between lake evaporation and ice cover is instead indicative of how winter evaporation is driven by overall winter meteorology, including but not dominated by ice condition. The only potential exceptions are two cases for Lake Superior in which the ice concentrations were greater than 90%. However, these likely represent an extreme case where the lake is exposed to low air temperatures and humidity when a mechanism for complete ice removal is unrealistic.

Analysis of specific months and years reveals that the increase in lake evaporation due to an ice-free state, that is, the difference between ice-covered and ice-free values, is less than the uncertainty in monthly evaporation derived from water balance methods when the ice cover is less than 50%. Although our simulations of intense, moderate, and mild winter years did not include months with ice cover between 50% and 90%, only at times with ice cover greater than 90% do we see that the impact of removing ice on evaporation exceeds the uncertainty from water balance estimates.

Overall, multi-decadal records and state-of-the-art turbulent flux algorithms reveal that ice cover itself does not serve as a strong cap on evaporation when considered at monthly and lake-wide scales. This is contrary to the typical narrative surrounding the implications of mild winters on Great Lakes hydrology and storage. While these lakes represent relatively diverse ice conditions and morphometry, it is important to consider that other large ice-covered lakes might experience different conditions, particularly overlake meteorology, that could impact evaporation-ice relationships. As such, these relationships should be considered for future projections of global freshwater storage and water resources management of Earth's largest ice-covered lakes.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All data sets used in this study are publicly available. The L2SWBM evaporation data are available from the CCGLBHHD (Coordinating Committee on Great Lakes Basic Hydraulic and Hydrologic Data, 2024). Ice concentration data and the surface water temperature (GLSEA) are available from the NOAA Coastwatch Program (NOAA CoastWatch, 2024). The GLCFS data are available from NOAA (NOAA Great Lakes Environmental Research Laboratory, 2024).

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